Documenting a network experiment

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why is “the model was fast and accurate” insufficient as an experiment record?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Record a network experiment with enough detail to replay its calculations. Try the recall before opening notes.

Your learning target

I can record a network experiment with enough detail to **replay** its calculations.

I can trace reported metrics to data roles and saved parameters.

I can write a bounded conclusion and identify missing evaluation **evidence**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Record a network experiment with enough detail to **replay** its calculations

Trace reported metrics to data roles and saved parameters

Write a bounded conclusion and identify missing evaluation **evidence**

Retrieve what you already know

> Why is “the model was fast and accurate” insufficient as an experiment record?

My starting explanation

This week’s end goal

By the end, I can explain documenting a network experiment using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Documenting a network experiment case. Separate observations, given facts and assumptions before choosing an explanation.

A useful experiment record explains what was asked, what was done, what happened and what remains unknown.

Use run NN32-A: does **fitting** one positive ReLU weight reduce validation mean squared error compared with its initial value?

This tiny constructed regression task teaches record keeping; it cannot establish performance on real-world data.

Data role and ID	Input x	Target y
Training T1	1	2
Training T2	2	4
Validation V1	3	6

The network predicts max(0, a*x), with one weight a and no bias. Start at a = 1, use learning rate 0.1, two full-batch **training** updates and mean squared error, the mean of squared **prediction** errors.

There is no normalisation, random initialisation or shuffling. Use Python float arithmetic and record the actual Python version when **replay**ing.

Before running, set the selection rule: choose the **checkpoint** with the smaller validation MSE, breaking a tie toward the earlier update. The initial model is the baseline.

Validation V1 chooses a **checkpoint** but never contributes to the **training** gradient. Final evaluation has not been run; its result must be recorded as not measured.

List the given facts and one fact that is not supplied.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Documenting a network experiment. Connect each labelled input, deciding step and result.

For these two **training** cases, MSE = 2.5*(a - 2) squared and its gradient is 5*(a - 2). At a = 1 the gradient is -5, so the first update is 1 - 0.1*(-5) = 1.5. The next gradient is -2.5, giving a = 1.75.

All preactivations in this run stay positive, so ReLU's derivative is one; this is not a general backpropagation implementation.

```
train = [(1.0, 2.0), (2.0, 4.0)]
validation = [(3.0, 6.0)]

def predict(a, x):
    return max(0.0, a*x)

def mse(a, rows):
    return sum((predict(a, x)-y)**2 for x, y in rows) / len(rows)

a, rate = 1.0, 0.1
baseline = mse(a, validation)
records = []
for update in [1, 2]:
    gradient = sum(2*(a*x-y)*x for x, y in train) / len(train)
    a -= rate * gradient
    records.append((update, a, mse(a, train), mse(a, validation)))
```

```
chosen = min(records, key=lambda r: (r[3], r[0]))
print(baseline, records, chosen[0])
```

The code assumes the supplied nonempty numeric datasets and positive preactivations. Each record stores update number, weight, **training** MSE and validation MSE after the update.

Keep both records, the initial weight and the rule; saving only the best score erases how the choice was made.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Documenting a network experiment. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Calculate both weight updates and give each **training** and validation MSE. Name the selected **checkpoint** and baseline validation MSE. How many independent validation examples support the comparison?

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: A teammate calls V4 a final test after using it to select a=1.25.

Predicted result

Prediction: Correct its role to validation; it was used for selection.

Why the result follows

Reason: Changing the label on a dataset does not make its **evidence** untouched.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Documenting a network experiment to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Run identifier	A unique label linking one experiment’s settings, outputs and records
Checkpoint	Saved model parameters at a specified stage of training
Manifest	An inventory connecting named artifacts with versions, roles and integrity information
Replay	Repeating a recorded procedure to check whether its outputs agree

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Documenting a network experiment, predict a step, then compare with its explanation.

Read each checkpoint against the same validation case

CHECKPOINT WEIGHT	TRAINING MSE	VALIDATION MSE
Update 1: a = 1.5	0.625	2.25
Update 2: a = 1.75	0.15625	0.5625

The baseline validation MSE is 9. The predeclared rule selects update 2 with validation MSE 0.5625 on one validation case. This is not a final-test score.

Worked explanation · Documenting a network experiment

A defensible conclusion is: in run NN32-A, update 2 reduced MSE on the single constructed validation case from 9 to 0.5625 compared with the initial model.

This **checkpoint** was selected using that same validation case; generalisation is untested. No timing, peak-memory or energy measurements are provided.

Using the selected weight, calculate the prediction for V1 and its squared error. Explain why the result must be labelled validation rather than independent final evaluation.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Documenting a network experiment

Document **checkpoint** selection so another team can reconstruct every reported number.

The facts we will use

Supplied facts

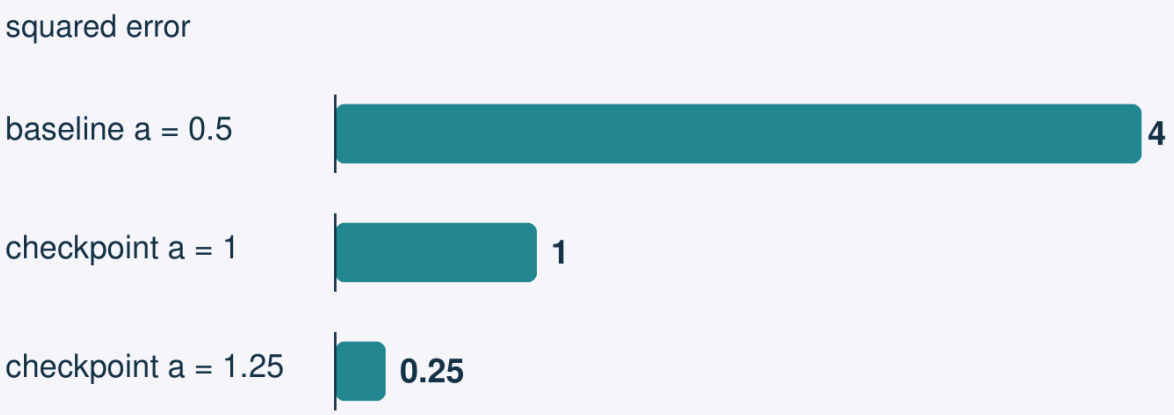
Run R8 predicts ReLU(a×x). Validation V4 has x = 2, target 3. Baseline a = 0.5; **checkpoint**s a = 1 and 1.25; choose minimum validation squared error.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/32

Validation squared loss by checkpoint



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/32

Documenting a network experiment

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Baseline prediction

At a = 0.5, prediction 1, error2 = 4.



2. Checkpoint comparison

At a = 1, prediction 2, loss 1; at 1.25, prediction 2.5, loss 0.25.



3. Select and record

Choose a = 1.25; save R8, inputs, formula, checkpoint values, loss rule and validation role.

CHANGE ONE THING

A teammate calls V4 a final test after using it to select a=1.25.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Baseline prediction	At a = 0.5, prediction 1, error² = 4.
2. Checkpoint comparison	At a = 1, prediction 2, loss 1; at 1.25, prediction 2.5, loss 0.25.
3. Select and record	Choose a = 1.25; save R8, inputs, formula, checkpoint values, loss rule and validation role.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A reproducible record includes the actual **input**, formula, parameters and scoring rule, together with each dataset's role. The validation case selects the best supplied **checkpoint**, so it is part of development.

Save that selection history and freeze the **checkpoint** before evaluating fresh final cases. Renaming the same case a test does not change how it was used.

What this example does not establish

Keep the stated limit

Reconstructing one run verifies arithmetic and provenance, not population performance or

educational learning impact.

Keep the stated limit

Read the labelled picture: What additional **evidence** is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: A teammate calls V4 a final test after using it to select a=1.25. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Documenting a network experiment claim and explain a repair using the supplied evidence.

Check the claim against the facts

Keep a **manifest** for the data with IDs and split membership, the exact code, settings, every **checkpoint**, raw **prediction** rows and metric calculations.

Add run ID, author/date, environment versions, known data provenance and permitted use, failed attempts, decisions and limitations.

A descriptive filename helps navigation, but does not prove that file contents stayed unchanged.

Check the claim against the facts

When saving artifacts, record their content hashes and keep the exact versions used. Hashes help detect content changes; they do not prove label correctness or experiment quality.

Retain the **input** order, any **training**-fitted pre**process**ing, model architecture and numeric precision.

Randomised experiments must also record seeds and the relevant random operations; a seed alone does not guarantee identical results across environments.

Check the claim against the facts

Ask a partner to **replay** from the record rather than from memory. If results differ, check **input**s, split membership, pre**process**ing, configuration, code, parameters and environment.

Document the discrepancy and the repair instead of overwriting the old record. A rerun on the same data checks the procedure; it does not add new independent evaluation cases.

Locate the earliest unsupported step

A folder contains only best-model.json and a screenshot of 0.5625. Identify four missing artifacts or records and explain what a content hash can and cannot verify.

State one repair and the checks you would rerun.

Practise together

> Practise Documenting a network experiment with support, then explain the deciding step in your own words.

For NN32-A, record the **prediction** rule, initial weight, bias policy, learning rate, update count, loss, **training**/validation IDs, data order or shuffling policy, and **checkpoint** selection rule.

Name one environment field still to record when you run it.

> **Using the selected weight, calculate the prediction for V1 and its squared error. Explain why the result must be labelled validation rather than independent final evaluation.**

> **Explain your prediction to a partner before comparing results.**

Practise with less help: Trace the selected result

Use workbook W2 for the connected example “Documenting a network experiment”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Documenting a network experiment. Proposed explanation: “A smaller MSE is an accuracy percentage”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What additional **evidence** is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Documenting a network experiment. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

> Calculate both weight updates and give each **training** and validation MSE. Name the selected **checkpoint** and baseline validation MSE. How many independent validation

examples support the comparison?

```
train = [(1.0, 2.0), (2.0, 4.0)]
validation = [(3.0, 6.0)]

def predict(a, x):
    return max(0.0, a*x)

def mse(a, rows):
    return sum((predict(a, x)-y)**2 for x, y in rows) / len(rows)

a, rate = 1.0, 0.1
baseline = mse(a, validation)
records = []
for update in [1, 2]:
    gradient = sum(2*(a*x-y)*x for x, y in train) / len(train)
    a -= rate * gradient
    records.append((update, a, mse(a, train), mse(a, validation)))

chosen = min(records, key=lambda r: (r[3], r[0]))
print(baseline, records, chosen[0])
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Documenting a network experiment. Explain why each step is justified.

Explain → apply → challenge

For NN32-A, record the **prediction** rule, initial weight, bias policy, learning rate, update count, loss, **training**/validation IDs, data order or shuffling policy, and **checkpoint** selection rule.

Name one environment field still to record when you run it.

A folder contains only best-model.json and a screenshot of 0.5625. Identify four missing artifacts or records and explain what a content hash can and cannot verify.

A partner gets different results after changing pre**process**ing and replacing the validation file. They keep the old run ID and overwrite the old score.

Explain what should be preserved and how the revised work should be recorded.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled documenting a network experiment **evidence** sheet to Image-pattern Network Investigation. Show how you can record a network experiment with enough detail to **replay** its calculations.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain

> practice; use the separately named independent case before feedback.

Report missing measurements as not measured, not as zero. If a final set has influenced a change, document that use and reserve new untouched evaluation before making an independent performance claim.

A polished report cannot fix leakage or missing **evidence**. For larger experiments, preserve denominators, subgroup results and resource conditions from Weeks 29 to 31.

> **A report says “final accuracy 0.5625, low energy” but includes only the selected weight and no data or code. Repair its claim and name the minimum evidence needed to check this run.**

Finish with a next action that follows from the limitations, such as planning a broader reserved evaluation without claiming it has already succeeded.

Next week begins the capstone by defining a hypothesis and baseline before controlled experiments.

A partner gets different results after changing pre**process**ing and replacing the validation file. They keep the old run ID and overwrite the old score.

Explain what should be preserved and how the revised work should be recorded.

> **My limited claim and a useful next check**

Study a complete case

> Revisit the complete Documenting a network experiment case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Read each checkpoint against the same validation case

CHECKPOINT WEIGHT	TRAINING MSE	VALIDATION MSE
Update 1: a = 1.5	0.625	2.25
Update 2: a = 1.75	0.15625	0.5625

The baseline validation MSE is 9. The predeclared rule selects update 2 with validation MSE 0.5625 on one validation case. This is not a final-test score.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Audit a new record independently

A separate run R17 records baseline a = 1 and **checkpoint**s a = 1.5 and a = 1.75. It uses **prediction** max(0, a*x) and one validation case V9 with x = 4, y = 8.

Compute baseline and **checkpoint prediction**s and squared errors, choose by lowest validation MSE, and write a bounded conclusion. Final evaluation and runtime have not been measured.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A separate run R17 records baseline a = 1 and **checkpoint**s a = 1.5 and a = 1.75. It uses **prediction** max(0, a*x) and one validation case V9 with x = 4, y = 8.

Compute baseline and **checkpoint prediction**s and squared errors, choose by lowest validation MSE, and write a bounded conclusion. Final evaluation and runtime have not been measured.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Documenting a network experiment, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Document **checkpoint** selection so another team can reconstruct every reported number.

By the end, I can	How I will show it
Record a network experiment with enough detail to replay its calculations	Explain the deciding step, show a trace or test, and state one limit.
Trace reported metrics to data roles and saved parameters	Explain the deciding step, show a trace or test, and state one limit.
Write a bounded conclusion and identify missing evaluation evidence	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Documenting a network experiment without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Record a network experiment with enough detail to **replay** its calculations

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Document **checkpoint** selection so another team can reconstruct every reported number.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role.

1. **Checkpoint** comparison

2. Select and record

3. Baseline **prediction**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Select and record Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role.

A reproducible record includes the actual **input**, formula, parameters and scoring rule, together with each dataset's role. The validation case selects the best supplied **checkpoint**, so it is part of development.

Save that selection history and freeze the **checkpoint** before evaluating fresh final cases. Renaming the same case a test does not change how it was used.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A teammate calls V4 a final test after using it to select a=1.25. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Correct its role to validation; it was used for selection.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Correct its role to validation; it was used for selection. Changing the label on a dataset does not make its **evidence** untouched.

Repair a missing building block

Review Efficiency and resource measurement (ai-5/31). Goal: Calculate throughput and amortised time per example from repeated timings. Short practice: Why is 10 ms/case not necessarily a single-request latency?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

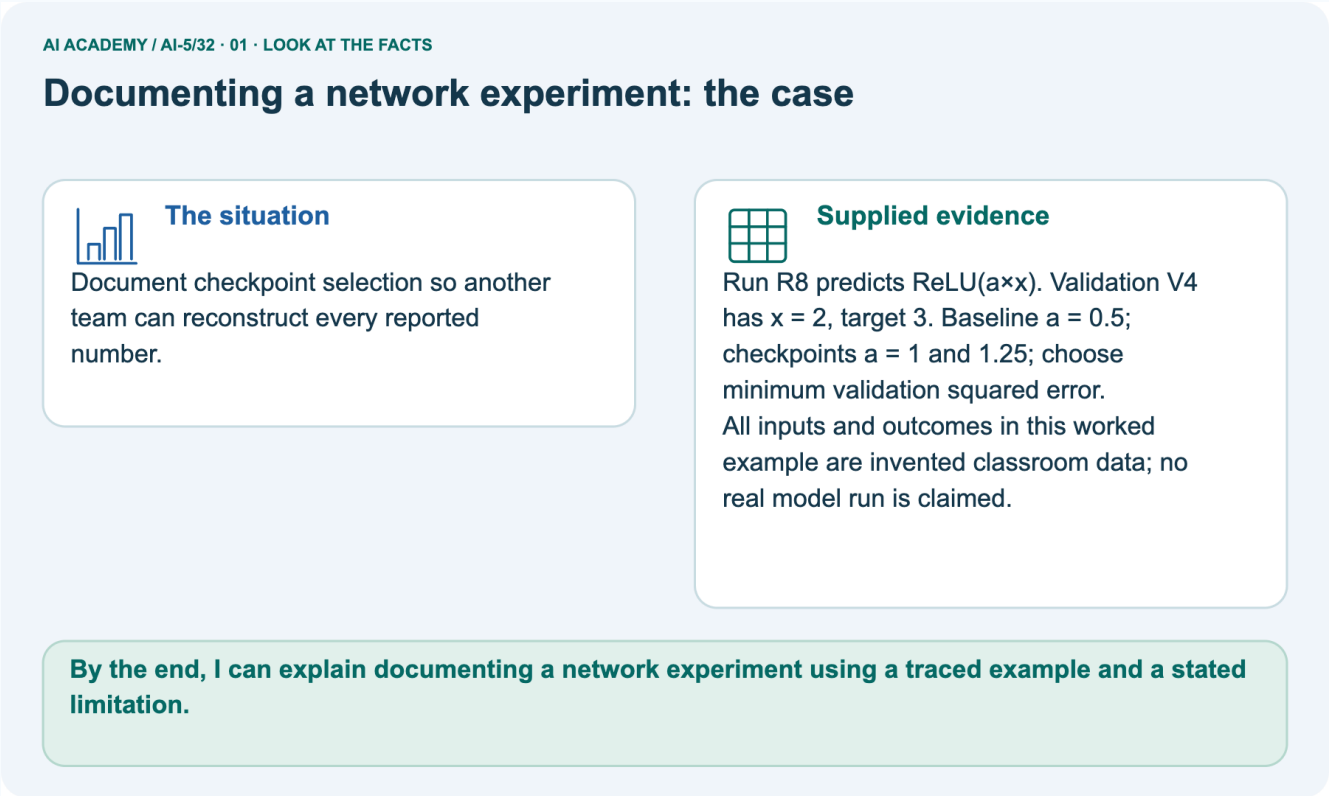
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Documenting a network experiment graphic using its labels, deciding evidence and relevant limitation.

01. Documenting a network experiment: the case



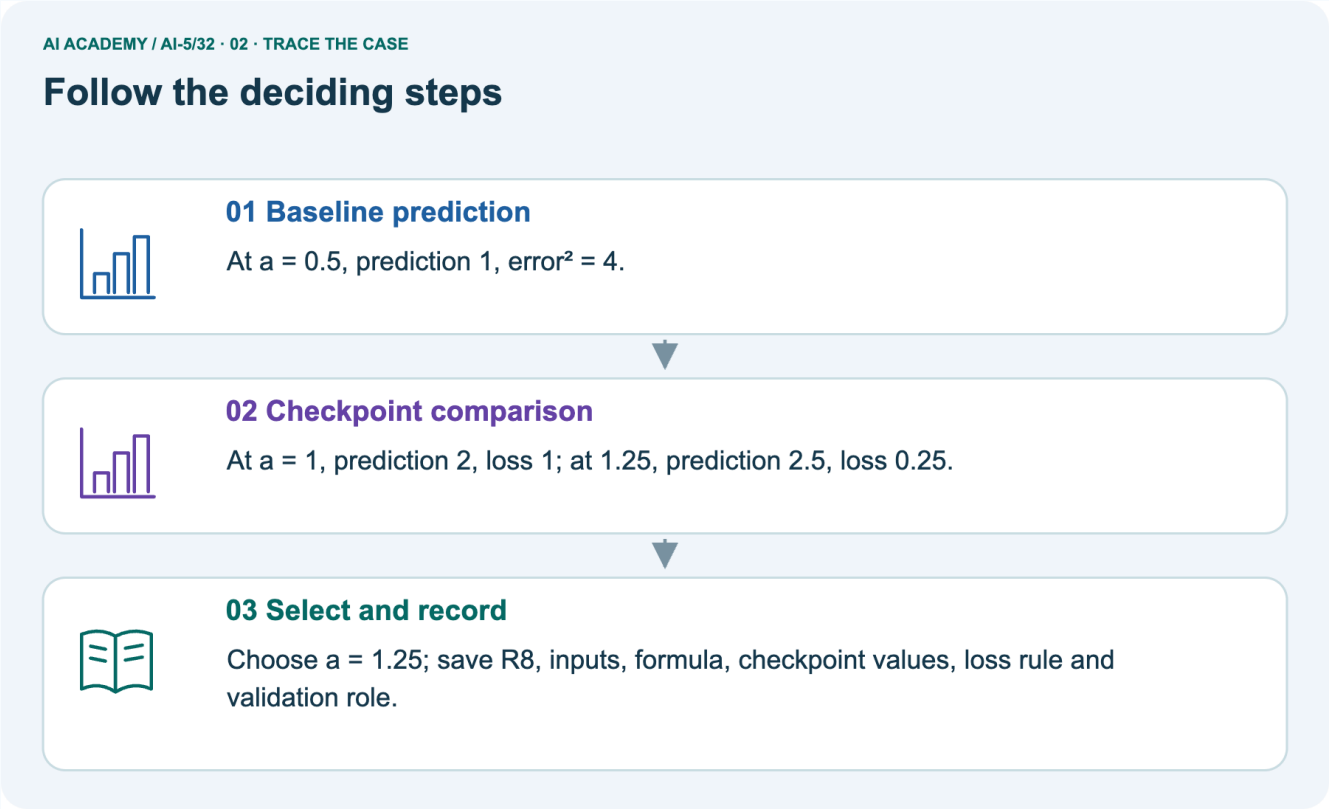
Follow the labelled picture with the matching explanation. Open the vector diagram

Document **checkpoint** selection so another team can reconstruct every reported number. Supplied facts: Run R8 predicts ReLU(a×x). Validation V4 has x = 2, target 3.

Baseline a = 0.5; **checkpoint**s a = 1 and 1.25; choose minimum validation squared error. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

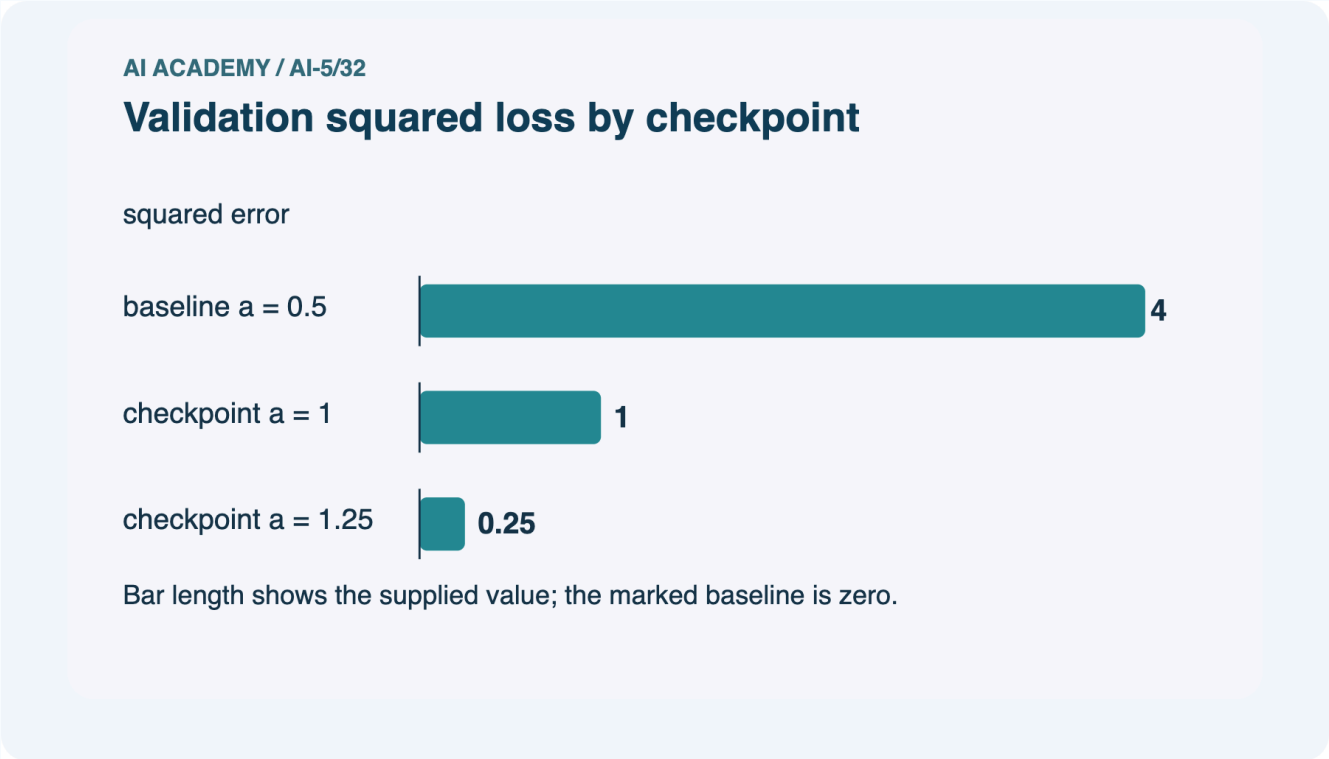


Follow the labelled picture with the matching explanation. Open the vector diagram

Baseline **prediction**: At a = 0.5, **prediction** 1, error² = 4. → **Checkpoint** comparison: At a = 1, **prediction** 2, loss 1; at 1.25, **prediction** 2.5, loss 0.25. → Select and record: Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Validation squared loss by checkpoint

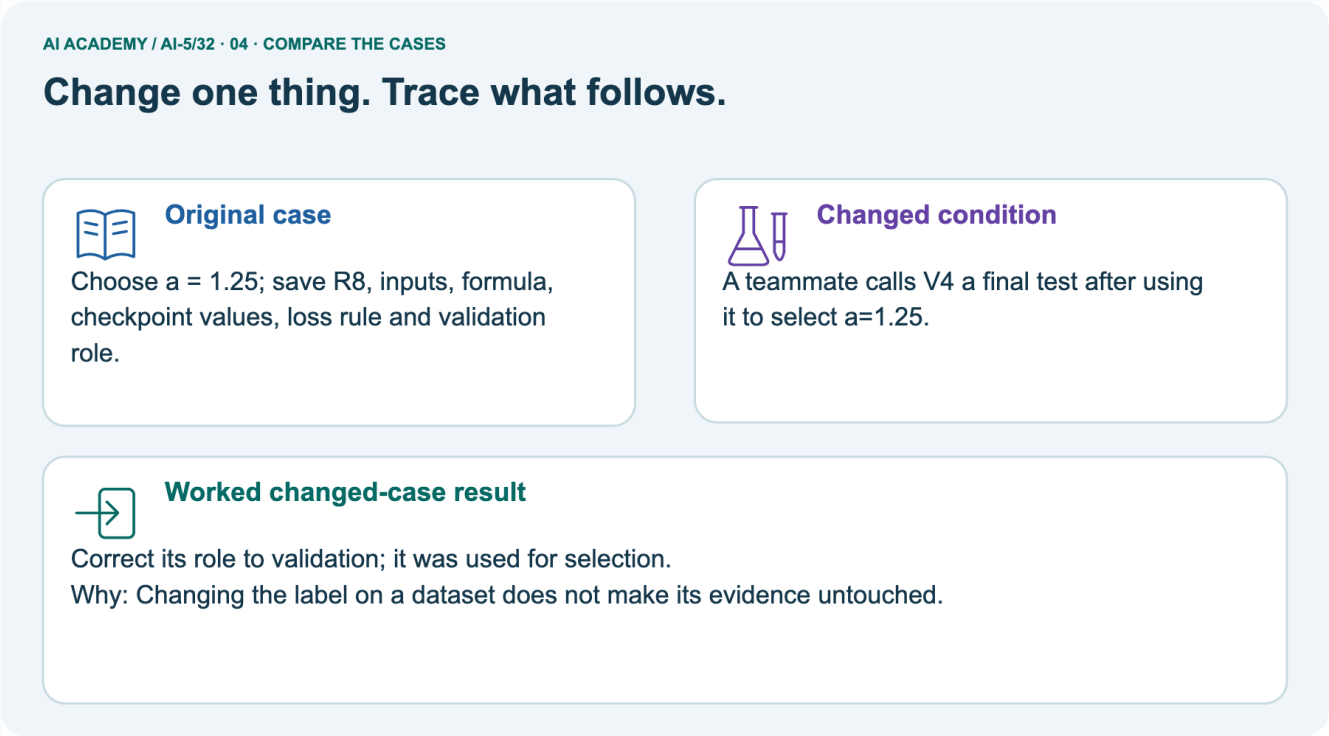


Follow the labelled picture with the matching explanation. Open the vector diagram

Validation squared loss by **checkpoint**. Units: squared error. baseline a = 0.5: 4. **checkpoint** a = 1: 1. **checkpoint** a = 1.25: 0.25.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



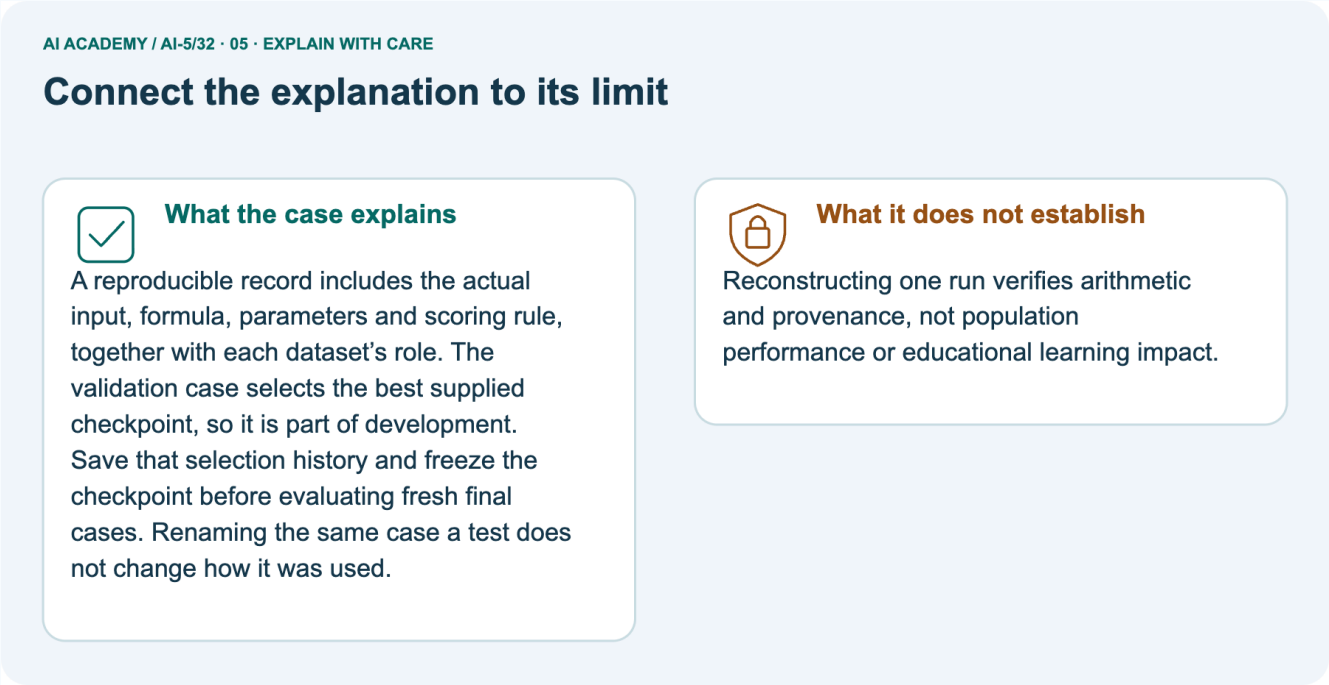
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role. Changed condition: A teammate calls V4 a final test after using it to select a=1.25.

Expected result: Correct its role to validation; it was used for selection. Reason: Changing the label on a dataset does not make its **evidence** untouched.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

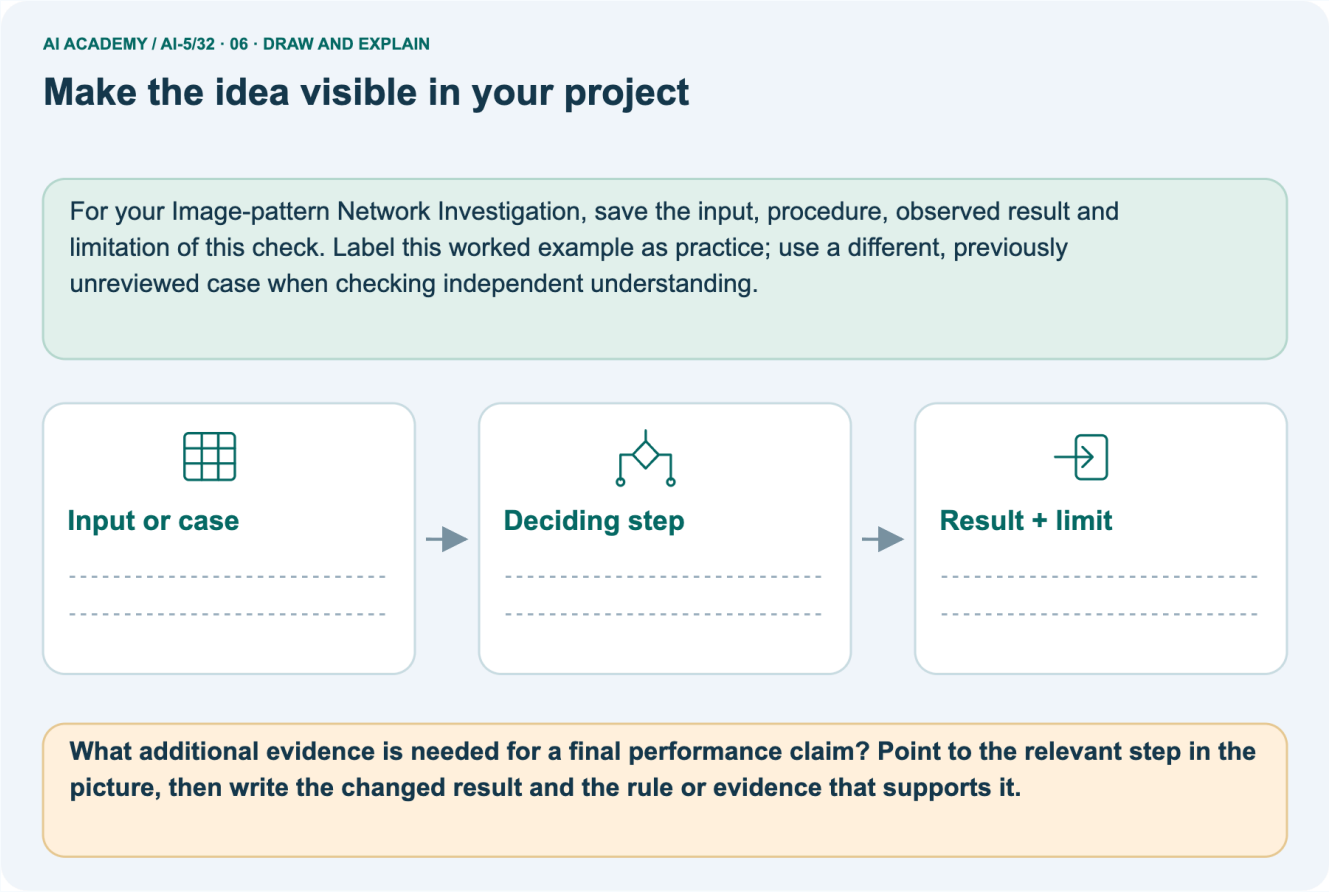
A reproducible record includes the actual **input**, formula, parameters and scoring rule, together with each dataset's role. The validation case selects the best supplied **checkpoint**, so it is part of development.

Save that selection history and freeze the **checkpoint** before evaluating fresh final cases. Renaming the same case a test does not change how it was used.

Limit: Reconstructing one run verifies arithmetic and provenance, not population performance or educational learning impact.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What additional **evidence** is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.